LSTM

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Assignment

Ans-1) $x_t = 0.5$, $h_{t-1} = 0.2$, $C_{t-1} = 0.4$

Forget gate:

$$f_t = \sigma(w_f x_t + U_f h_{t-1} + b_f)$$

$$= \sigma(0.7(0.5) + 0.3(0.2) + 0.1)$$

$$= \sigma(0.51)$$

$$= 0.6248$$

Input gate: $i_t = \sigma(w_i x_t + U_i h_{t-1} + b_i)$

$$= \sigma(0.6(0.5) + 0.4(0.2) + 0)$$

$$= 0.5939$$

Candidate Cell State =

$$\tilde{C}_t = \tanh(w_c x_t + U_c h_{t-1} + b_c)$$

$$= \tanh(0.45 + 0.04)$$

$$= 0.4542$$

New Cell State:-

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t$$

$$C_t = 0.6248(0.4) + (0.5939)(0.4542)$$

$$C_t = 0.5197$$

Output gate:-

$$O_t = \sigma(w_o x_t + U_o h_{t-1} + b_o)$$

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$= \sigma(0.5(0.5) + 0.5(0.2) + 0.1)$

$= \sigma(0.45) = 0.6106$

Hidden State (h_t) :-

$h_t = O_t \cdot \tanh(C_t)$

$\tanh(0.5197) \simeq 0.4775$

$h_t = (0.6106)(0.4775)$

$= 0.2916$

Q2 input dimension $n = 10$

Hidden unit $h = 32$

(a) Input weight $W = 32 \times 10 = 320$

$U = 32 \times 32 = 1024$

Bias $= 32$

$320 + 1024 + 32 = 1378$

Four gates $= 4 \times 1376 = 5504$

(b) Two bias per gate :-

$32 + 32 = 64$

$320 + 1024 + 32 + 32 = 1408 \rightarrow$ for 1 gate

for 4 gates $= 4(1408) = 5632$ parameter

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Q3 Ans f^k, where f = forget gate value, k = no of the timestamp

a) $f = 0.9$

After 10 steps $= (0.9)^{10} = 0.3487$

After 50 $= (0.9)^{50} = 0.0052$

b) After 50 steps

$f = 0.5$

$(0.5)^{50}$

$= 0.8818 \times 10^{-16}$

$= 0.000$

The gradient has disappeared

$f = 0.99$

$(0.99)^{50}$

$= 0.6050$

c) If the forget gate is small, the gradient becomes extremely small over time steps.

Therefore leaving information for 50 steps earlier becomes very difficult.

Q4 Given :-

→ 3 years

→ 365 days / years

→ Lookback = 168 hrs

→ Forecast = 24

→ Featurd = 3

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→ Batch size = 64

a) Number of hourly records
$3 \times 365 \times 24 = 26280$

b) Number of windows
Each sample requires = 168 + 24 = 192

For N records, stride 1
Windows = N - lookback - forecast
= 26089

c) Tensor shapes
= 168 time steps
= 3 features
One input (168, 3)
For ~~fetch~~ batch size 64:

= (64, 168, 3)

The model predicts 24 feature load values
= (64, 24)

d Training window & Mini-Batches
26280 (0.70)
= 18396
18396 - 168 - 24 + 1 = 18205
⇒ $\frac{18205}{64}$ = 285 mini batches

e) LSTM parameters

$x = 3$

$h = 64$

Single bias $= 4(hx + h^2 + h)$

$= 4(64(3) + 64^2 + 64)$

$= 17408$ parameters

Q5

a) Min-max normalization

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$$x' = \frac{95 - 20}{120 - 20} = 0.75$$

b) Predictions = Failure, $P \geq 0.5$

No failure, $P < 0.5$

$0.837 \geq 0.5 \rightarrow$ Failure

c) True Positive (TP) = 30

False Positive = Predicted failure − TP

$= 45 - 30 = 15$

False Negative = Actual Failure − TP

$40 - 30 = 10$

Non Failure = $1000 - 40 = 960$

$TN = \text{Actual non failure} - FP$

$= 960 - 15$

$= 945$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{30}{30+15} = 66.67\%$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{30}{30+10} = 75\%$$

$$F_1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= \frac{2 \times \left(\frac{30}{45}\right) \times \left(\frac{30}{40}\right)}{\frac{30}{45} + \frac{30}{40}} = 70.54\%$$

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

$$= \frac{30+945}{1000} = 97.50\%$$

d) Recall should be prioritized because missing an actual pump failure (false negative) can cause serious damage and cost. Accuracy can be misleading because failure are rare here only (40 out of 1000). A model can achieve high accuracy but missing imp. features